

Flag-Adapted Schur–Stein Metrics for Directional Hardy Packets

Exact Dissipation and the Outer-Packet Bottleneck

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Abstract

The directional small-noise program has reduced its remaining analytic gate to two Hilbert–Schmidt Hardy energies. RH-58 replaced oblique radial Riesz blocks by unitary Schur packets, but a scalar sum over all upper-triangular paths grew much faster than the exact energies. This paper retains the Schur flag and replaces that path ledger by positive anisotropic metrics.

For every stable finite upper-triangular block matrix, we solve a canonical Lyapunov equation on each diagonal block and prove that hierarchical positive scalings produce a block diagonal metric P with $R = P - T^*PT \succ 0$. This gives an exact flag-compatible stability theorem. For an observation Y , the constant $\kappa = \|YR^{-1/2}\|_2^2$ makes κP a positive Stein supersolution. Applying a separate prefix metric to each initial Schur packet bounds every diagonal entry of the RH-58 packet Gram. The complete dissipation R is always at least as sharp as the usual one-number contraction estimate. A two-scalar-block formula also proves an endpoint tradeoff: when an outer packet is seen through a nonzero inner observation, the metric cost diverges at both boundaries of the admissible scaling interval.

The five-scale all-column binary64 audit gives a substantial but incomplete gain. At $\sigma = 0.01$, the left/right packetwise metric uppers are 19.2196 and 12.1382, compared with the RH-58 absolute-path uppers 1922.40 and 380.40 and exact energies 1.4681 and 1.7603. The remaining loss is concentrated in the outer packet, whose metric uppers are 13.7336 and 6.7965. The fitted total growth exponents are still 0.898 and 0.763; therefore no dyadically uniform or polylogarithmic Stage A1 estimate follows. A 256-bit Arb calculation certifies the formulas on one two-scalar-block model only. No arithmetic trace formula, self-adjoint realization, or Hilbert–Polya conclusion is claimed.

Keywords: Schur flag; Stein inequality; Lyapunov metric; Hardy energy; block diagonal stability; nonnormal operator; small noise.

MSC 2020: 47A10; 47B65; 93B07; 93D05; 65F35.

1 Introduction

RH-48–RH-50 reduced small-noise intrinsic Riesz identification to two directional Hardy energies of residue-deflated bulk operators [5, 6, 8]. The exact finite-dimensional energy is benign on every stored scale, but a proof must remain quantitative as the noise decreases. RH-56 ruled out a uniform strong-space route [7]; RH-57 showed that fixed radial Riesz blocks become severely oblique [9].

RH-58 then moved to a unitary Schur basis and proved two exact positive packet Gram identities [10]. That coordinate change removed the projector wall: the measured input and output packet budgets stayed near the exact Hardy energies. However, bounding every feed-forward Schur path independently produced smallest-scale uppers 1922.40 and 380.40. The path argument had discarded the Hilbert-space geometry that motivated the Schur basis.

The natural repair is a positive anisotropic metric. This idea has two logically separate parts.

1. Does every stable Schur flag admit a compatible block diagonal Lyapunov metric?
2. Can such metrics control the directional packet energies with a noise-uniform quantitative budget?

The answer to the first question is yes. We give an explicit construction from local Lyapunov equations and diagonal block scalings. The second answer is mixed. Exact dissipation removes most of the RH-58 path proliferation, but the outer packet develops a strong endpoint cost on the two finest levels. Thus RH-59 identifies a much narrower obstruction without closing Stage A1.

Contributions and boundary

1. We prove flag-compatible block diagonal stability for every finite stable Schur partition.
2. We convert the complete metric dissipation into a positive packetwise observability Stein supersolution.
3. We prove that exact dissipation dominates both the global contraction estimate and its scalar block-norm comparison.
4. We derive a two-block endpoint tradeoff and audit the resulting packet certificates on five all-column folded-Gaussian levels.

The matrix theorems are exact finite-dimensional statements. The production Schur forms, optimized scalings, fitted powers, and dense Lyapunov solves are binary64 diagnostics. No continuum regularity of the noise-dependent Schur flags or uniform control of their hierarchical weights is proved.

2 Directional Hardy packets

Let A be a stable finite matrix, X an input matrix, and Y an observation matrix. The Hardy energy is

$$\mathcal{E}^2 = \sum_{m \geq 0} \|Y A^m X\|_{\mathfrak{S}_2}^2 = \text{tr}(X^* O X), \quad O - A^* O A = Y^* Y. \quad (1)$$

The two RH-50 directions have this form after the radius rescaling $A = r^{-1}N$, with $r = 0.85$ in the audit.

Choose a unitary Schur partition [1, 2]

$$A = Q T Q^*, \quad T = \begin{pmatrix} D_1 & T_{12} & \cdots & T_{1J} \\ 0 & D_2 & \cdots & T_{2J} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & D_J \end{pmatrix}. \quad (2)$$

Write $\hat{X} = Q^* X$, $\hat{Y} = Y Q$, and let E_j be the orthogonal coordinate projections. The input packets are $X_j = E_j \hat{X}$. Their complete responses

$$F_j = (\hat{Y} T^m X_j)_{m \geq 0} \quad (3)$$

have Gram matrix $K_{ij} = \langle F_i, F_j \rangle$. In particular,

$$e_j^2 := K_{jj} = \text{tr}(X_j^* \hat{O} X_j), \quad \hat{O} = Q^* O Q. \quad (4)$$

RH-58 proved

$$\mathcal{E} \leq \sum_{j=1}^J e_j, \quad \mathcal{E}^2 \leq \theta \sum_{j=1}^J e_j^2, \quad (5)$$

where θ is the largest eigenvalue of the normalized packet Gram. The present task is to bound the diagonal packet energies without summing every triangular path absolutely.

Upper triangularity supplies the key localization. The prefix

$$\mathcal{H}^{(j)} = E_1 \mathcal{H} \oplus \cdots \oplus E_j \mathcal{H} \quad (6)$$

is invariant under T , and the orbit of X_j remains inside this prefix. We may therefore choose a different positive metric for each packet.

3 Canonical flag metrics

For every diagonal block, let H_i be the local Lyapunov metric [4, 11]

$$H_i - D_i^* H_i D_i = I, \quad H_i = \sum_{m \geq 0} (D_i^*)^m D_i^m \succ 0. \quad (7)$$

Define normalized couplings and their scalar comparison by

$$B_{ij} = H_i^{1/2} T_{ij} H_j^{-1/2}, \quad c_{ij} = \|B_{ij}\|_2 \quad (i \leq j). \quad (8)$$

The matrix $C = (c_{ij})$ is upper triangular.

Theorem 3.1 (Flag-compatible block metric). *Let T be a finite stable upper-triangular block matrix. For positive numbers $s_1, \dots, s_J$, put*

$$P_s = \bigoplus_{i=1}^J s_i^2 H_i, \quad S_s = P_s^{1/2} T P_s^{-1/2}. \quad (9)$$

Then

$$\|S_s\|_2 \leq \|\text{diag}(s_i) C \text{diag}(s_i)^{-1}\|_2. \quad (10)$$

Moreover,

$$c_{ii}^2 = 1 - \frac{1}{\lambda_{\max}(H_i)} < 1. \quad (11)$$

Consequently there exist positive scales s_i for which

$$P_s - T^* P_s T \succ 0. \quad (12)$$

The same assertion holds on every invariant prefix.

Proof. For a block vector $v = (v_j)$, put $u_j = \|v_j\|$. The i th block of $S_s v$ is bounded by

$$\sum_{j \geq i} \frac{s_i}{s_j} c_{ij} u_j.$$

Taking the Euclidean norm over i proves (10). From (7),

$$B_{ii}^* B_{ii} = I - H_i^{-1},$$

which gives (11).

Choose $0 < \varepsilon < 1$ and $s_i = \varepsilon^{J-i}$. Every strictly upper entry of the scaled comparison is multiplied by ε^{j-i} , while its diagonal is fixed. As $\varepsilon \downarrow 0$, its spectral norm tends to $\max_i c_{ii} < 1$. Hence (10) is strictly below one for all sufficiently small ε . Finally,

$$P_s - T^* P_s T = P_s^{1/2} (I - S_s^* S_s) P_s^{1/2} \succ 0.$$

Restricting the construction to the first j blocks proves the prefix statement. $\square$

Remark 3.2 (Existence is not a uniform estimate). The proof may require $s_1 \ll s_2 \ll \dots \ll s_J$. Such a hierarchy can make an inner observation expensive in the dual metric. The theorem solves finite-dimensional compatibility, not the small-noise endpoint budget.

4 Exact-dissipation Stein supersolutions

Fix packet j and work on its invariant prefix. Let P_s be any metric from Theorem 3.1 with

$$R_s = P_s - (T^{(j)})^* P_s T^{(j)} \succ 0. \quad (13)$$

Write $Y^{(j)}$ for the observation restricted to this prefix and define

$$\kappa_{s,j} = \left\| Y^{(j)} R_s^{-1/2} \right\|_2^2. \quad (14)$$

Theorem 4.1 (Packetwise exact-dissipation certificate). *The matrix $\tilde{O}_{s,j} = \kappa_{s,j} P_s$ satisfies*

$$\tilde{O}_{s,j} - (T^{(j)})^* \tilde{O}_{s,j} T^{(j)} \succeq (Y^{(j)})^* Y^{(j)}. \quad (15)$$

Therefore the exact packet energy obeys

$$\boxed{e_j^2 \leq u_{s,j}^2 := \kappa_{s,j} s_j^2 \left\| H_j^{1/2} X_j \right\|_{\mathfrak{S}_2}^2.} \quad (16)$$

Consequently

$$\mathcal{E} \leq \sum_{j=1}^J u_{s(j),j} \quad (17)$$

for arbitrary feasible packet-specific metrics. If an independent packet coherence bound θ is available, then also

$$\mathcal{E}^2 \leq \theta \sum_{j=1}^J u_{s(j),j}^2. \quad (18)$$

Proof. By the definition of operator norm,

$$(Y^{(j)})^* Y^{(j)} \preceq \kappa_{s,j} R_s,$$

which is exactly (15). Iterating the positive Stein residual and using stability gives $O^{(j)} \preceq \tilde{O}_{s,j}$. Testing this inequality on X_j proves (16). The triangle inequality for the complete packet responses gives (17); substituting $e_j \leq u_{s(j),j}$ into (5) gives (18). $\square$

The complete dissipation is sharper than reducing the metric to one contraction number.

Proposition 4.2 (Dissipation versus contraction). *Let $q_s = \left\| P_s^{1/2} T^{(j)} P_s^{-1/2} \right\|_2 < 1$. Then*

$$\kappa_{s,j} \leq \frac{\left\| Y^{(j)} P_s^{-1/2} \right\|_2^2}{1 - q_s^2}, \quad (19)$$

and q_s is bounded by the scalar comparison in (10).

Proof. The factorization in Theorem 3.1 gives $R_s \succeq (1 - q_s^2) P_s$. Inverting reverses the Loewner order and proves (19). The final statement is (10). $\square$

This distinction matters numerically. At many optimized production weights, the exact normalized matrix is contractive even though the scalar block-norm comparison exceeds one.

5 A two-block endpoint tradeoff

The remaining metric loss is not only an optimization artifact. It already appears in the real two-scalar-block model

$$T = \begin{pmatrix} a & \gamma \\ 0 & b \end{pmatrix}, \quad P = \text{diag}(p_1, p_2), \quad Y = (y_1, y_2), \quad (20)$$

where $|a|, |b| < 1$ and $\gamma \neq 0$.

Proposition 5.1 (Endpoint coercivity). *Put $\alpha = 1 - a^2$, $\delta = 1 - b^2$, and normalize $p_2 = 1$, $t = p_1/p_2$. Then $R = P - T^*PT$ is positive definite exactly when*

$$0 < t < \frac{\alpha\delta}{\gamma^2}. \quad (21)$$

On this interval the sharp Stein multiplier is

$$\kappa(t) = \frac{y_1^2(\delta - \gamma^2 t) + 2a\gamma y_1 y_2 t + \alpha y_2^2 t}{t(\alpha\delta - \gamma^2 t)}. \quad (22)$$

If $y_1 \neq 0$ and $a\gamma y_1 + \alpha y_2 \neq 0$, then

$$\kappa(t) \longrightarrow +\infty \quad \text{at both endpoints of (21)}. \quad (23)$$

Hence an outer source packet has a finite interior optimal scaling, but no extreme hierarchy removes its inner-observation cost.

Proof. Direct multiplication gives

$$R = \begin{pmatrix} \alpha t & -a\gamma t \\ -a\gamma t & \delta - \gamma^2 t \end{pmatrix}, \quad \det R = t(\alpha\delta - \gamma^2 t).$$

Sylvester's criterion proves (21). Since Y has one row, $\kappa = YR^{-1}Y^*$, which gives (22). The lower-end divergence follows from $y_1 \neq 0$. At the upper endpoint the numerator vanishes only in the excluded alignment $a\gamma y_1 + \alpha y_2 = 0$. This proves (23). $\square$

6 Five-scale deterministic audit

6.1 Construction

We use exactly the RH-58 all-column folded-Gaussian family

$$(\sigma, N) = (0.16, 32), (0.08, 64), (0.04, 128), (0.02, 256), (0.01, 512), \quad N\sigma = 5.12,$$

the Hardy radius $r = 0.85$, and physical radial cuts

$$0.15, \quad 0.35, \quad 0.55. \quad (24)$$

Empty Schur bands are omitted. For every active diagonal block we solve (7) densely. For packet j we fix $s_j = 1$, first find a feasible seed from the scalar comparison, and then minimize the exact binary64 value in (16) over the remaining logarithmic scales. The optimizer only proposes weights. Acceptance requires direct positive dissipation with minimum eigenvalue above 10^{-8} and a checked Stein supersolution residual.

The exact observability Gramian is solved independently and supplies the packet diagonals used for comparison. Every optimizer terminated successfully. None of these floating-point checks is an interval proof.

6.2 Route comparison

σ	exact $\mathcal{E}$		RH-59 metric upper		RH-58 path upper	
	left	right	left	right	left	right
0.16	0.9040	1.0026	1.1719	1.5183	1.59	1.78
0.08	1.1626	1.2653	2.1206	2.4307	4.41	4.15
0.04	1.3338	1.4845	2.8953	4.0798	22.39	14.26
0.02	1.4096	1.6340	3.9855	7.5263	137.26	71.18
0.01	1.4681	1.7603	19.2196	12.1382	1922.40	380.40

Table 1: Exact Hardy energies, packetwise absolute synthesis (17), and the inherited scalar path upper. The last four columns are sufficient upper bounds, not lower bounds.

At the smallest scale, exact dissipation improves the old path upper by factors 100.0 and 31.3. The finite-range growth exponents fall from 2.54 and 1.96 for scalar paths to 0.898 and 0.763 for the flag metrics. This is a genuine route improvement, but the remaining exponents are not compatible with the desired polylogarithmic conclusion.

The measured packet coherence constants at $\sigma = 0.01$ are 1.171 and 1.099. If inserted into (18), they give diagnostic uppers 15.286 and 7.837. These values are not standalone certificates because no analytic uniform coherence theorem is supplied here.

6.3 The outer-packet bottleneck

packet	left			right		
	exact	metric	ratio	exact	metric	ratio
central	1.1768	1.1814	1.00	1.2311	1.4109	1.15
inner cloud	0.4731	1.7510	3.70	0.5129	2.0147	3.93
middle cloud	0.4168	2.5536	6.13	0.4811	1.9161	3.98
edge cloud	0.5262	13.7336	26.10	1.1310	6.7965	6.01

Table 2: Exact packet energies and optimized flag-metric uppers at $\sigma = 0.01$. The loss accumulates toward the outer edge packet.

The central left certificate is within 0.4% of its exact packet energy. The edge packet is qualitatively different. Its fitted upper exponents are 1.178 and 1.009, although its exact energies remain order one. On the left, the optimized normalized contraction reaches 0.993861 and the smallest normalized dissipation eigenvalue is 0.01224. The optimizer is balancing close-to-singular dissipation against a rapidly growing dual observation cost, precisely the mechanism of Proposition 5.1.

Across all cases, the largest local metric condition number is 9.86, the largest local Lyapunov residual is 5.6×10^{-16} , the largest global observability Stein residual is 4.6×10^{-15} , and the largest Schur reconstruction defect is 7.1×10^{-15} . The most negative computed supersolution eigenvalue is -9.2×10^{-13} , consistent with binary64 roundoff at a theoretically semidefinite boundary.

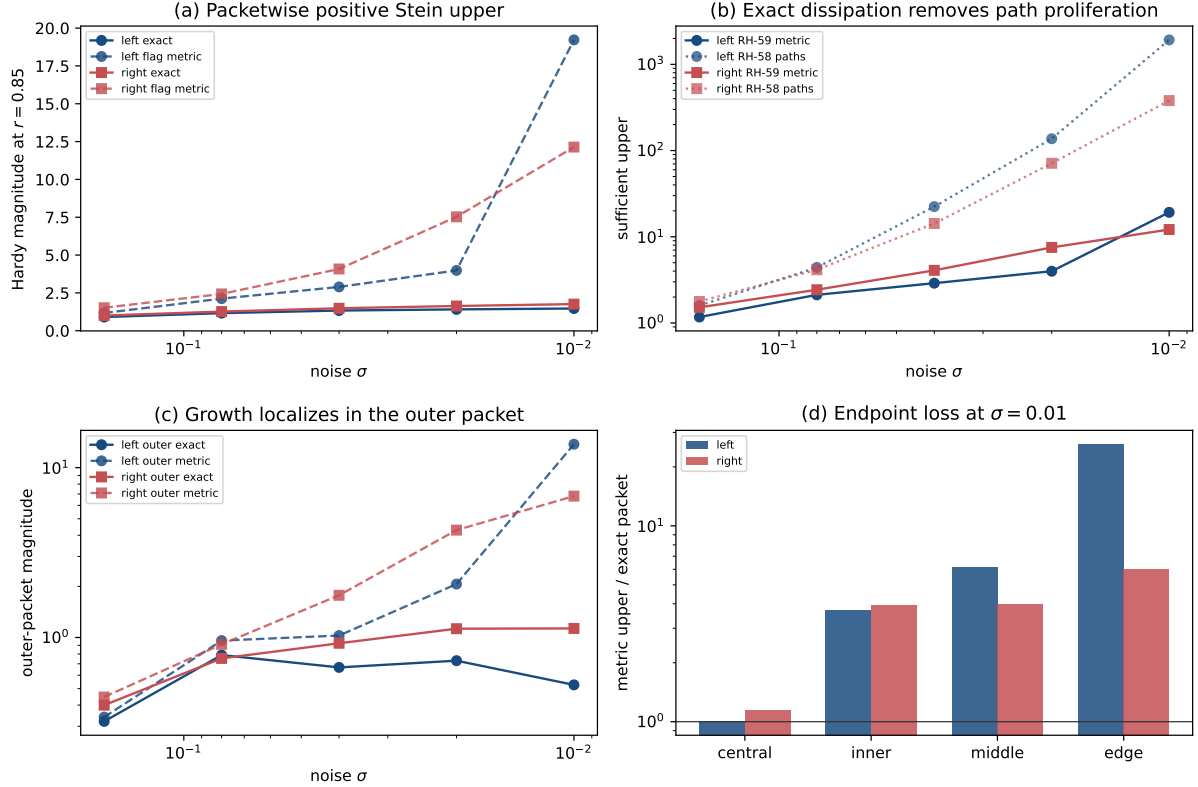


Figure 1: Flag-metric audit. (a) Packetwise positive Stein uppers versus the exact energies. (b) Exact dissipation removes most of the RH-58 path proliferation. (c) Remaining growth localizes in the outer packet. (d) The smallest-scale endpoint loss increases along the Schur flag.

6.4 Outward-rounded model audit

A 256-bit Arb calculation [3] evaluates (20) at

$$a = 0.2, \quad b = 0.7, \quad \gamma = 0.3, \quad Y = (0.8, 0.6), \quad (s_1, s_2) = (0.25, 1).$$

It encloses both local Lyapunov identities, proves $R \succ 0$, inflates the sharp multiplier by $1 + 10^{-6}$, verifies the resulting Stein supersolution by strictly positive principal minors, and certifies the outer-packet upper. This validates the formulas on the displayed scalar model only; no production Schur metric is interval enclosed.

7 Program consequence

Flag-metric existence	Closed for every stable finite Schur partition. Diagonal-block compatibility is not the missing theorem.
Scalar absolute paths	Replaced. Exact positive dissipation reduces the smallest-scale bounds by factors 100.0 and 31.3.
Central packet	Quantitatively mild; its local metric certificate is close to exact on the stored family.
Outer packet	Still obstructed. Hierarchical scaling transfers the cost into the dual inner observation, and the fitted upper grows with a positive power.

Stage A1	Open. Neither the metric absolute synthesis nor the empirical coherence-assisted variant supplies a polylogarithmic theorem.
Stage A4	Still conditional on Stage A1; the earlier factor and cutoff interfaces are unchanged.

The next target is therefore more specific than after RH-58. A viable argument must control the outer-to-inner observation channel without paying for a nearly singular hierarchical metric. Possibilities include a frequency-localized observability inequality, a multiscale physical packet basis whose outer injection is already aligned with the central channel, or a phase-aware estimate for the off-diagonal packet Gram. Reoptimizing the same static block weights is unlikely to remove the endpoint tradeoff.

8 No arithmetic or Hilbert–Polya conclusion

Nothing here constructs a self-adjoint operator, a $T \log T$ counting law, a von Mangoldt or prime-power trace formula, or a zeta-zero identity. No Riemann-hypothesis conclusion is drawn. The independent twin-prime branch is not used.

9 Reproducibility

The archive contains the flag-metric algebra, theorem tests, five-scale all-column results, Arb model certificate, figures, dependency hashes, and publication artifacts. Principal commands are

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_flag_metric_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_arb_flag_metric_audit.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The flag-compatible metric, exact-dissipation supersolution, dominance inequality, packet synthesis, and two-block tradeoff are analytic finite-dimensional results. All production asymptotics remain numerical. Stage A1, unconditional intrinsic identification, and every arithmetic or Hilbert–Polya conclusion remain outside the claims.

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